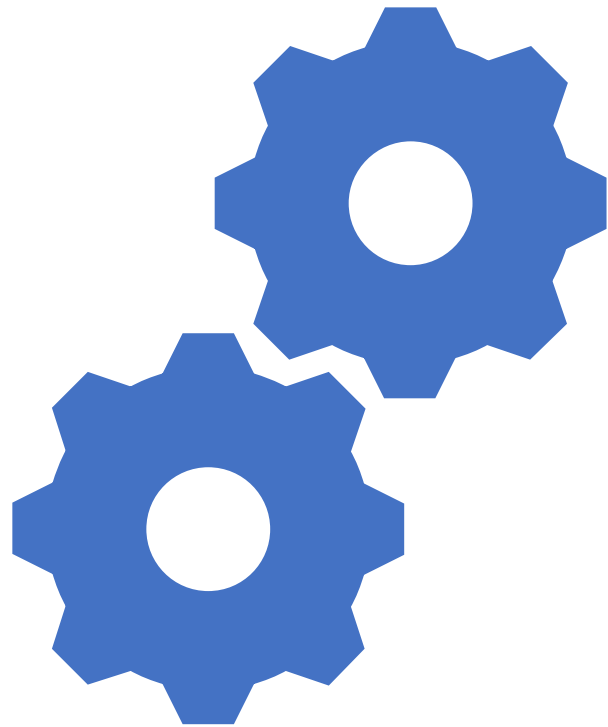




Program using loops and nested loops

Half Pyramid of *

```
#include <stdio.h>
int main() {
    int i, j, rows;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; ++i) {
        for (j = 1; j <= i; ++j) {
            printf("*\t");
        }
        printf("\n");
    }
    return 0;
}
```

```
*
*      *
*      *      *
*      *      *      *
*      *      *      *      *
```

Inverted half pyramid of *

```
#include <stdio.h>
int main() {
    int i, j, rows;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = rows; i >= 1; --i) {
        for (j = 1; j <= i; ++j) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

```
* * * * *
* * * *
* * *
* *
*
```

Full Pyramid of *

```
#include <stdio.h>

int main() {
    int i, space, rows, k = 0;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; ++i, k = 0) {
        for (space = 1; space <= rows - i; ++space) {
            printf(" ");
        }
        while (k != 2 * i - 1) {
            printf("* ");
            ++k;
        }
        printf("\n");
    }
    return 0;
}
```

```
      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * *
```

Full Pyramid of Numbers

```
  1
 2 3 2
3 4 5 4 3
4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

```
#include <stdio.h>
int main() {
    int i, space, rows, k = 0, count = 0, count1 = 0;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; ++i) {
        for (space = 1; space <= rows - i; ++space) {
            printf(" ");
            ++count;
        }
        while (k != 2 * i - 1) {
            if (count <= rows - 1) {
                printf("%d ", i + k);
                ++count;
            } else {
                ++count1;
                printf("%d ", (i + k - 2 * count1));
            }
            ++k;
        }
        count1 = count = k = 0;
        printf("\n");
    }
    return 0;
}
```

Inverted full pyramid of *

```
#include <stdio.h>
int main() {
    int rows, i, j, space;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = rows; i >= 1; --i) {
        for (space = 0; space < rows - i; ++space)
            printf(" ");
        for (j = i; j <= 2 * i - 1; ++j)
            printf("* ");
        for (j = 0; j < i - 1; ++j)
            printf("* ");
        printf("\n");
    }
    return 0;
}
```

```
* * * * *
* * * * *
* * * * *
* * *
*
```

Pascal's Triangle

```

    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

```
#include <stdio.h>
int main() {
    int rows, coef = 1, space, i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 0; i < rows; i++) {
        for (space = 1; space <= rows - i;
space++)
            printf(" ");
        for (j = 0; j <= i; j++) {
            if (j == 0 || i == 0)
                coef = 1;
            else
                coef = coef * (i - j + 1) / j;
            printf("%4d", coef);
        }
        printf("\n");
    }
    return 0;
}
```

Floyd's Triangle

```
1
2 3
4 5 6
7 8 9 10
```

```
#include <stdio.h>
int main() {
    int rows, i, j, number = 1;
    printf("Enter the number of
rows: ");
    scanf("%d", &rows);
    for (i = 1; i <= rows; i++) {
        for (j = 1; j <= i; ++j) {
            printf("%d ", number);
            ++number;
        }
        printf("\n");
    }
    return 0;
}
```


Program to Check Palindrome

```
#include <stdio.h>
int main() {
    int n, reversed = 0, remainder, original;
    printf("Enter an integer: ");
    scanf("%d", &n);
    original = n;

    // reversed integer is stored in reversed variable
    while (n != 0) {
        remainder = n % 10;
        reversed = reversed * 10 + remainder;
        n /= 10;
    }

    // palindrome if original and reversed are equal
    if (original == reversed)
        printf("%d is a palindrome.", original);
    else
        printf("%d is not a palindrome.", original);

    return 0;
}
```

C Program to Display Fibonacci Sequence

The Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21

```
// print 3rd to nth terms
for (i = 3; i <= n; ++i) {
    printf("%d, ", nextTerm);
    t1 = t2;
    t2 = nextTerm;
    nextTerm = t1 + t2;
}

return 0;
}
```

```
#include <stdio.h>
int main() {
```

```
    int i, n;
```

```
    // initialize first and second terms
    int t1 = 0, t2 = 1;
```

```
    // initialize the next term (3rd term)
    int nextTerm = t1 + t2;
```

```
    // get no. of terms from user
    printf("Enter the number of terms: ");
    scanf("%d", &n);
```

```
    // print the first two terms t1 and t2
    printf("Fibonacci Series: %d, %d, ", t1, t2);
```



Thank you